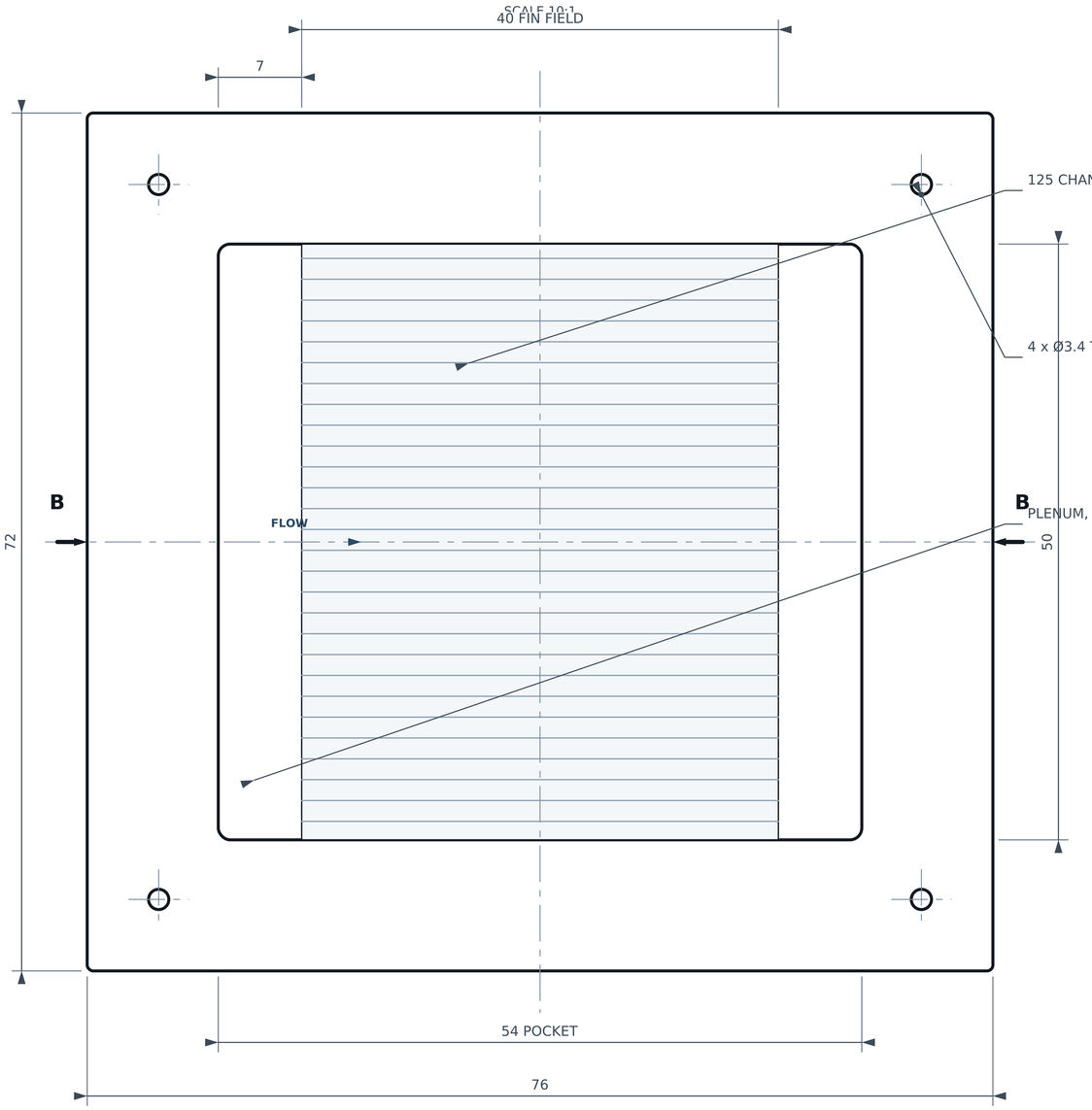
PLAN — FIN SIDE UP

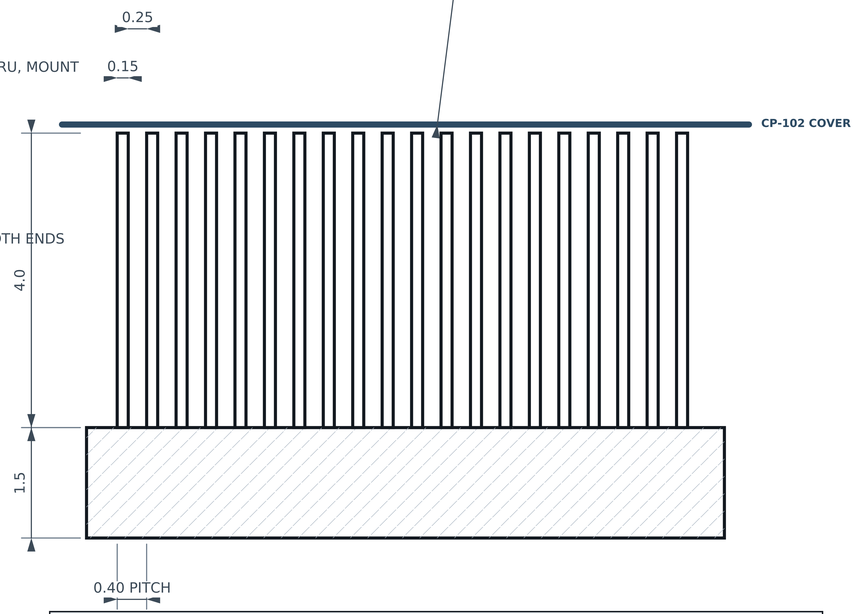
SCALE 10:1



DETAIL E — FIN ROOT

TIP CLEARANCE 0.05 MAX — NOT BRAZED

SCALE 60:1



SECTION B—B (ALONG FLOW)

SCALE 10:1

NOTES

1. MATL: OXYGEN-FREE copper C10200 / JIS C1020.
2. Fins skived from solid. No brazed or bonded fin.
3. Braze land (perimeter, 8 wide) flat 0.02, Ra 0.8.
4. Die-side face flat 0.015, Ra 0.4, no burr.
5. Deburr channel ends. A raised burr blocks a channel.
6. Flush and dry before braze. No cutting fluid in the fins.
7. Handle fin field with a cover plate. Bent fins are scrap.

CHANNEL PERFORMANCE (PG25)	
channels	125
hydraulic diameter	0.471 mm
velocity @ 1.5 L/min	0.20 m/s
Reynolds	71 laminar
h	5300 W/m2K
fin efficiency	0.55
pressure drop	6.1 kPa

Aspect ratio 16:1. Skiving is the only economical way to cut 0.15 mm fins this tall in copper.

CRITICAL CHARACTERISTICS			
FEATURE	**NOMINAL**	**TOL**	**CRITICAL TO**
Base thickness under die	1.5	+0.05 / -0.00	conduction, 3.3 K
Fin height	4.0	±0.05	flow area / dP
Channel width	0.25	±0.02	dP goes as w^-2
Die-side flatness	—	0.015	TIM bond line
Braze land flatness	—	0.02	hermetic joint
Channel blockage	zero	100 % flow test	one blocked channel = hot spot

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **COLD PLATE BODY**
Cu C10200 (JIS C1020) / 76 x 72 x 6.0 / 125 SKIVED FINS

SCALE: **10:1** — UNITS: **mm** — THIRD ANGLE PROJECTION

DRAWING No. **CP-101** — REV **A** — SHEET **2 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-10